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(54) **METHOD, APPARATUS, AND SYSTEM FOR COMMON MODE FEEDBACK CIRCUIT USING SWITCHED CAPACITORS**

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(52) **U.S. Cl.** **330/9**; 327/554; 330/69; 330/258

(58) **Field of Search** 330/9, 69, 258; 327/554

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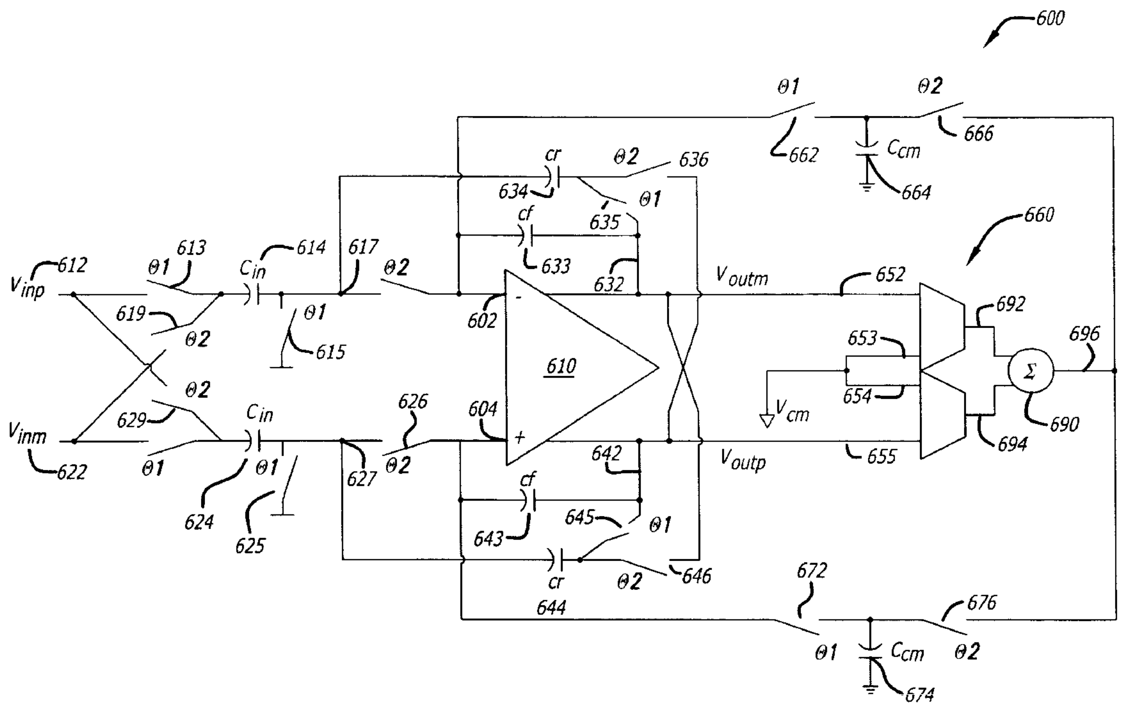
(57) **ABSTRACT**

According to one aspect of the invention, an apparatus is provided which includes a first amplifier and a common mode feedback circuit. The first amplifier includes a first input, a second input, a first output, and a second output. The first input and the second input are coupled to receive a first input voltage and a second input voltage, respectively. The first and second outputs provide a first output voltage and a second output voltage, respectively. The common mode feedback circuit has first and second switched capacitors coupled to provide a common mode feedback signal based on the first and second output voltages to the first and second inputs of first amplifier.

26 Claims, 4 Drawing Sheets



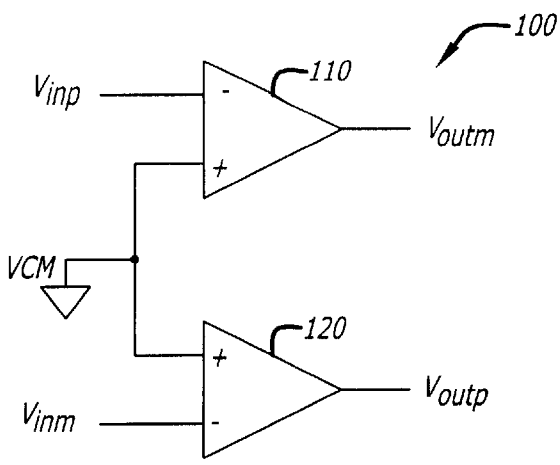


FIG. 1
(Prior Art)

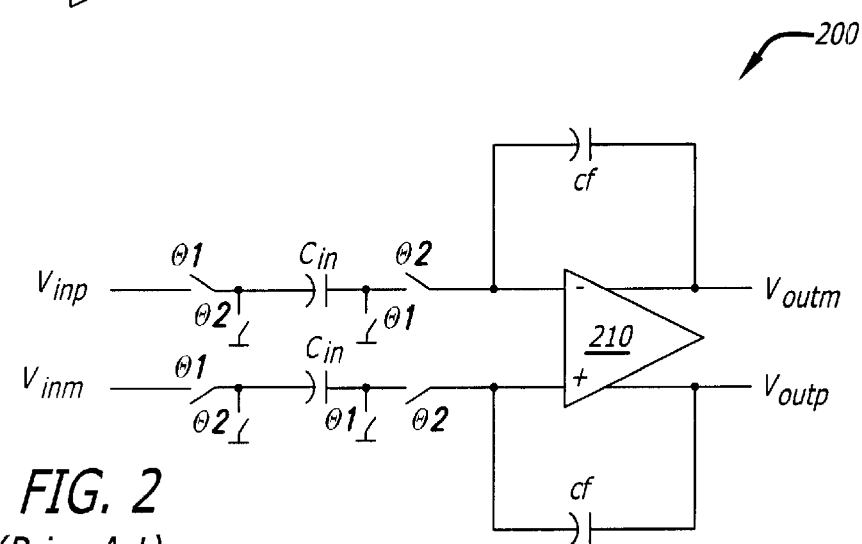


FIG. 2
(Prior Art)

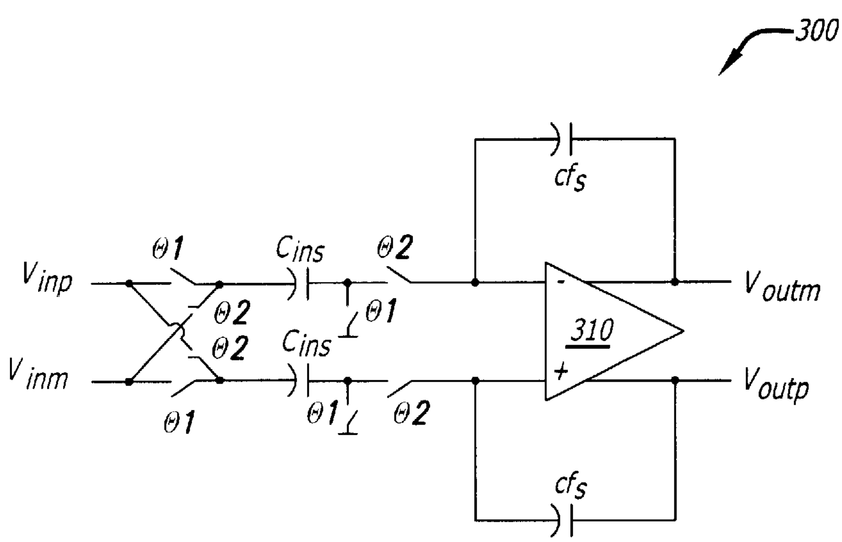


FIG. 3
(Prior Art)

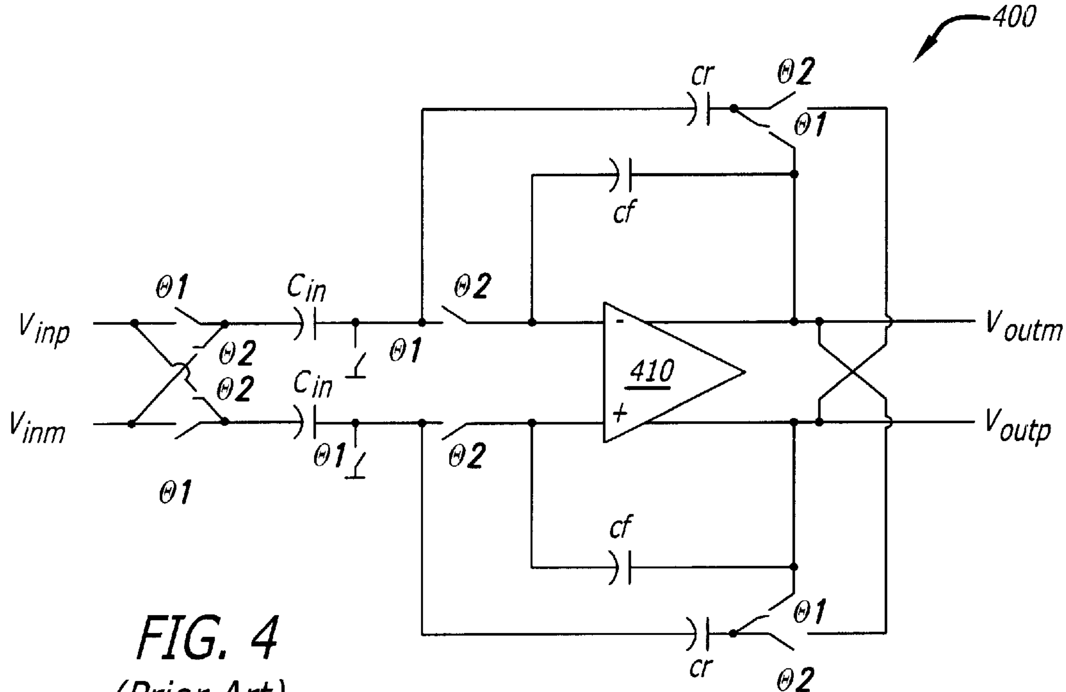


FIG. 4
(Prior Art)

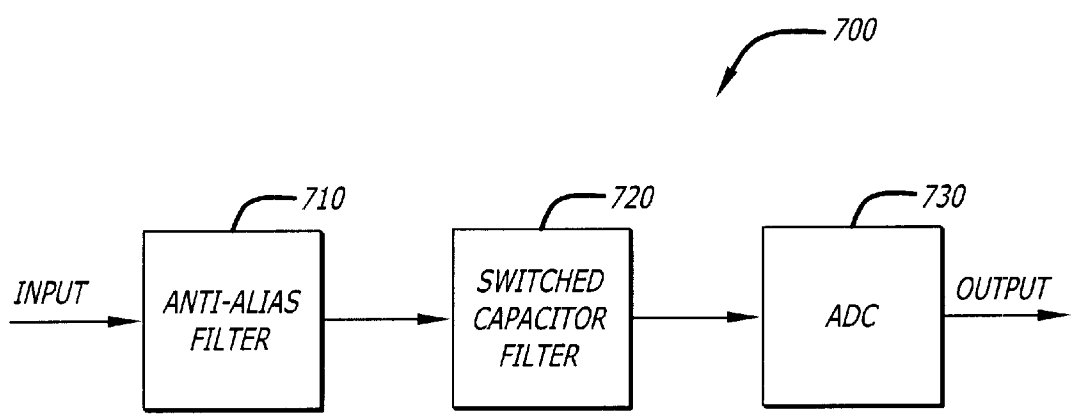


FIG. 7

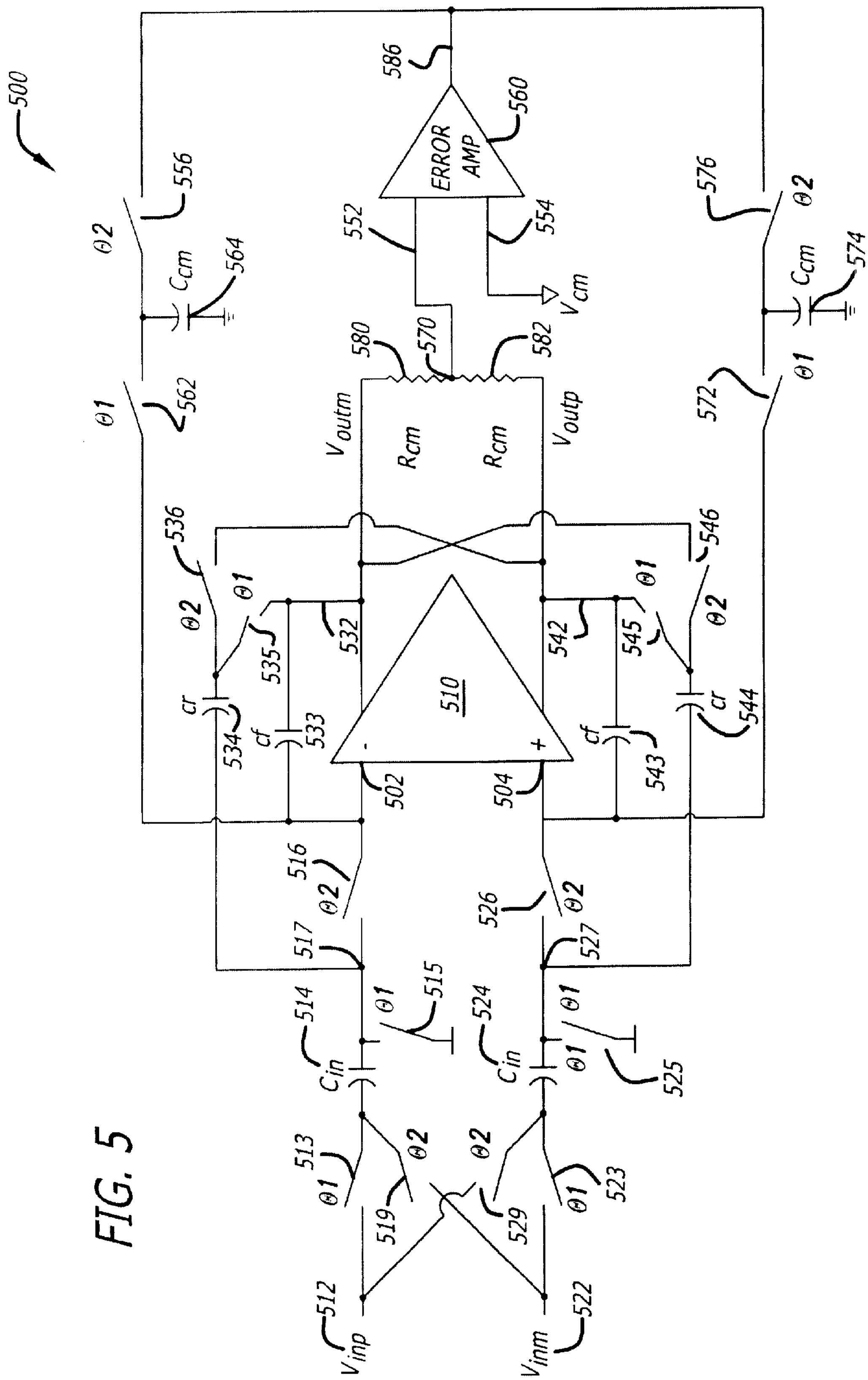
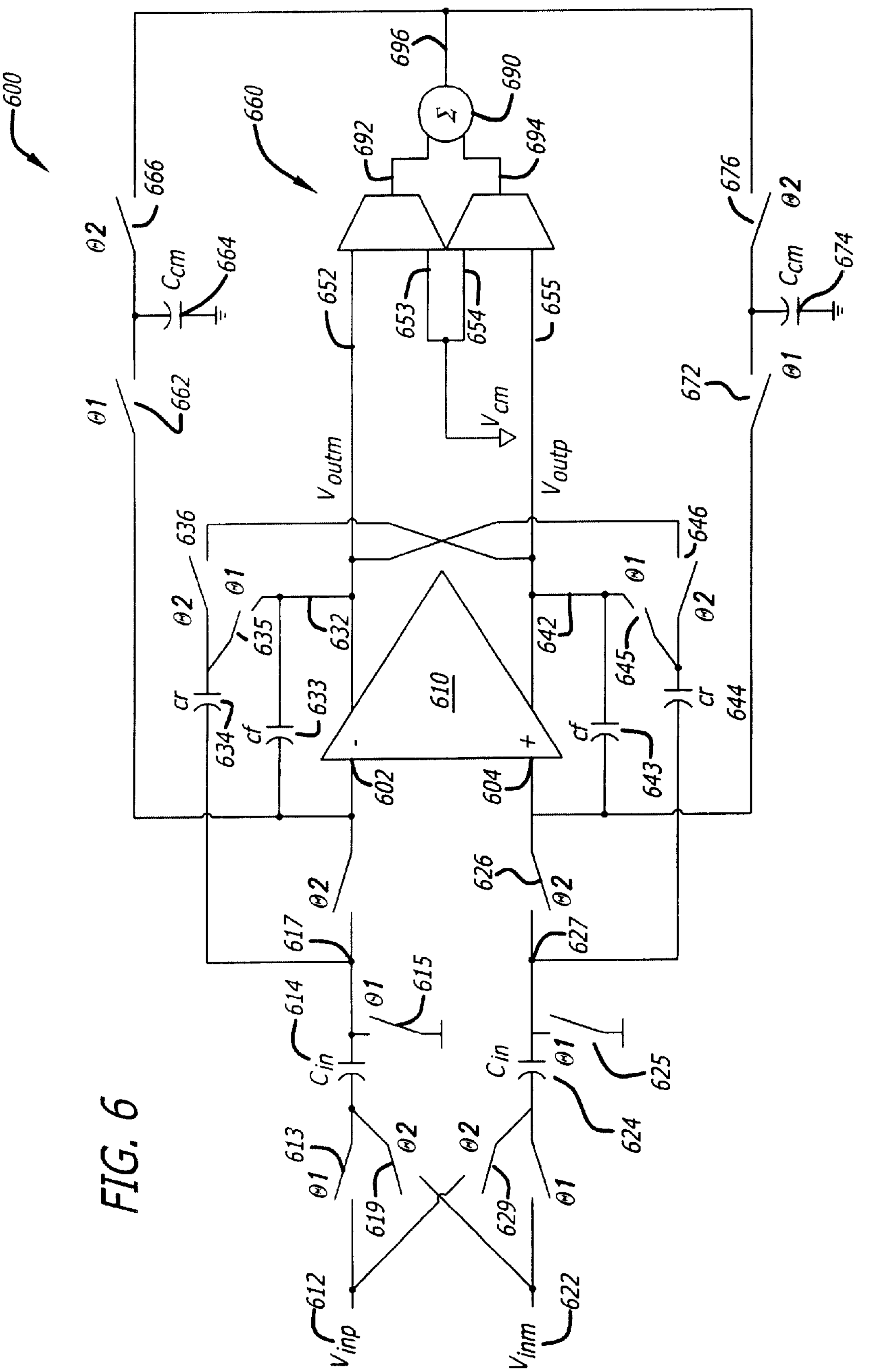

500
ERROR AMP
560
586
552
554
V_{cm}
556
562
564
C_{cm}
$\theta 1$
$\theta 2$
576
572
574
580
570
582
R_{cm}
V_{outm}
V_{outp}
510
502
504
532
533
534
535
536
542
543
544
545
546
cr
cf
512
513
514
515
516
517
519
522
523
524
525
526
527
529
C_{in}
V_{inp}
V_{inm}

FIG. 5


600
660
610
696
690
692
694
666
676
664
674
662
672
652
653
654
655
C_{cm}
$\Theta 1$
$\Theta 2$
V_{cm}
V_{outm}
V_{outp}
636
646
632
633
634
635
642
643
644
645
cr
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602
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617
627
626
614
615
624
625
613
619
629
612
622
C_{in}
V_{inp}
V_{inm}

FIG. 6

METHOD, APPARATUS, AND SYSTEM FOR COMMON MODE FEEDBACK CIRCUIT USING SWITCHED CAPACITORS

FIELD OF THE INVENTION

The present invention relates to the field of switched capacitor filters design and implementation. More specifically, the present invention relates to a common mode feedback circuit using switched capacitors.

BACKGROUND OF THE INVENTION

In many filter designs, a differential amplifier can be used to improve the signal to noise ratio (SNR) and to improve immunity to noise coupling into the circuit. One of the drawbacks to differential amplifiers is the need for a common-mode feedback circuit. In order for the common-mode loop to be stable, the poles introduced in the common-mode loop need to be well above those of the differential loop. This can require large amounts of power and design complexity. A solution to this problem is to use two single-ended amplifiers with their positive inputs tied to the desired common-mode voltage. Generally, this circuit has the same properties as the fully differential design with respect to the SNR and noise immunity improvement without having to use a common-mode feedback structure.

One of the limitations on power consumption in switched capacitor circuits is the need to quickly charge switched capacitors to a desired accuracy in a limited period of time (e.g., half a clock cycle). Since the transfer function in a switched capacitor filter is based on capacitor ratios, the capacitors could be made as small as the photo-lithography and parasitic effects will permit without altering the transfer function. The true limit on capacitor sizes however, arises from the noise resulting from the resistive switch that is sampled onto the capacitor along with the desired signal. This noise power is inversely proportional to the switched capacitor value, hence for a required noise performance, a minimum capacitor size is indicated. This will also determine the capacitive load on the amplifiers in the switched-capacitor filter. For a given clock frequency and charge accuracy, the power required by the operational amplifiers (opamps) approximately scales with the square of the capacitor size. One way to reduce the capacitor size without degrading the noise performance is to use cross-sampling. Cross-sampling is a method where the switched capacitor is referenced to the inverse of the signal rather than an analog ground, effectively doubling the amount of charge transferred by the capacitor allowing the same transfer function to be realized with one-half the capacitance.

However, there is a problem in using the pseudo-differential and the cross-sampling methods in switched capacitor filter design because there is no common-mode information being transferred from the input to the output of the opamp. The common mode transfer function is not defined since there is no path for the input common mode to the output. In addition, there is no feedback of the output common mode voltage. In the presence of offsets, the output of the filter will saturate and the filter will no longer perform its intended function.

BRIEF DESCRIPTION OF THE DRAWINGS

The features of the present invention will be more fully understood by reference to the accompanying drawings, in which:

FIG. **1** shows a circuit diagram of a pseudo-differential amplifier used in filter design;

FIG. **2** illustrates a circuit diagram of a typical switched capacitor integrator;

FIG. **3** shows a circuit diagram of a typical switched capacitor integrator with cross-sampling;

FIG. **4** shows a circuit diagram of a typical switched capacitor low pass filter with cross-sampling;

FIG. **5** shows a circuit diagram of one embodiment of a switched capacitor filter with common mode feedback, according to the teachings of the present invention;

FIG. **6** illustrates a circuit diagram of another embodiment of a switched capacitor filter with common mode feedback, in accordance with the teachings of the present invention; and

FIG. **7** shows a block diagram of a system **700** in which the teachings of the present invention are implemented.

DETAILED DESCRIPTION

In the following detailed description numerous specific details are set forth in order to provide a thorough understanding of the present invention. However, it will be appreciated by one skilled in the art that the present invention may be understood and practiced without these specific details.

FIG. **1** shows a circuit diagram of a pseudo-differential amplifier **100** used in filter designs. As shown in FIG. **1**, the pseudo-differential amplifier **100** includes two single-ended amplifiers **110** and **120** with their positive inputs tied to a desired common-mode voltage VCM. As mentioned above, this type of pseudo-differential amplifier circuit basically has the same properties as the fully differential design with respect to the SNR and noise immunity improvement without having to use a common-mode feedback structure.

FIG. **2** illustrates a circuit diagram of a typical switched capacitor integrator **200**. As shown in FIG. **2**, the two inputs Vinp and Vinm are coupled to the two respective input terminals of the opamp **210** via the respective switches Θ_1, the respective capacitors Cin and the respective switches Θ_2. The respective input terminals of the opamp **210** are coupled to the respective output terminals Voutm and Voutp, respectively, via the capacitors Cf.

The noise generated in the on resistance of the switches is sampled onto the capacitors Cin and then transferred to the integrating capacitors Cf. The noise and signal charges transferred on each clock phase are given below:

$$Q_{N,1}=\text{Noise charge sampled on } C_{IN} \text{ during } \Theta_1=C_{IN}V_N=C_{IN}*\mathrm{sqrt}(kT/C_{IN})=\mathrm{sqrt}(kTC_{IN}) \quad \text{(Eq. 1)}$$

where V_N is the noise voltage, k is the bozemans constant, T is the temperature (in Kelvin).

$$Q_{N,2}=\mathrm{sqrt}(kTC_{IN}) \quad \text{(Eq. 2)}$$

$$Q_{SIG,1}=\text{Signal charge sampled on } C_{IN} \text{ during } \Theta_1=C_{IN}*V_{IN} \quad \text{(Eq. 3)}$$

All of these charges are transferred to the integrating capacitor, but since the two noise charges are uncorrelated, their noise powers will add, not their noise charge. The output noise voltage and signal-to-noise ratio (SNR) is given below:

$$V_{OUT}=(C_{IN}V_{IN}+\mathrm{sqrt}(2kTC_{IN}))/C_F \quad \text{(Eq. 4)}$$

$$SNR=(C_{IN}V_{IN}/C_F)/\mathrm{sqrt}(2kTC_{IN})/C_F)=\mathrm{sqrt}(C_{IN}/2kT) \quad \text{(Eq. 5)}$$

FIG. **3** shows a circuit diagram of a typical switched capacitor integrator with cross-sampling **300**. The switched capacitor integrator **300** includes an opamp **310** having two input terminals and two output terminals. As described above, cross-sampling is one way to reduce the capacitor size in a switched capacitor filter without degrading the noise performance. In this cross-sampling configuration, the noise and signal charges transferred to the integrating capacitors are given below:

$$Q_{N,1}=\text{Noise charge sampled on } C_{IN} \text{ during } \Theta_1=C_{INS}V_N=C_{IN}*\text{sqrt}(kT/C_{INS})=\text{sqrt}(kTC_{INS}) \quad \text{(Eq. 6)}$$

$$Q_{N,2}=\text{sqrt}(kTC_{INS}) \quad \text{(Eq. 7)}$$

$$Q_{SIG,1}=\text{Signal charge sampled on } C_{IN} \text{ during } \Theta_1=C_{INS}*V_{IN} \quad \text{(Eq. 8)}$$

$$Q_{SIG,2}=\text{Signal charge sampled on } C_{IN} \text{ during } \Theta_2=C_{INS}*V_{IN} \quad \text{(Eq. 9)}$$

As discussed above, the noise charges are uncorrelated, so their noise power, rather than their noise voltage, will add. The output voltage Vout and the SNR are given below:

$$V_{OUT}=(2C_{INS}V_{IN}+\text{sqrt}(2kTC_{INS}))/C_{FS} \quad \text{(Eq. 10)}$$

$$SNR=(2C_{INS}V_{IN}/C_{FS})/\text{sqrt}(2kTC_{INS})/C_{FS})=\text{sqrt}(4C_{INS}/2kT) \quad \text{(Eq. 11)}$$

It can be observed based upon the noise analysis of the switched capacitor integrators described above that the size of the switched capacitors may be reduced by a factor of four and the size of the integrating capacitors may be reduced by a factor of two, without degrading the signal-to-noise ratio (SNR).

As mentioned above, there exists a problem in using the pseudo-differential amplifier and the switched capacitor integrator with cross-sampling in switched capacitor design.

FIG. **4** shows a circuit diagram of a typical switched capacitor low pass filter **400** with cross-sampling. The low pass filter **400** includes an opamp **410** having two input terminals and two output terminals. As illustrated in FIG. **4**, there is no common-mode information transferred from the input to the output of the low pass filter. Typically, the z-domain transfer function of the filter is shown in the following equations:

$$V_{OUT,\ DIFF}=V_{OUTP}-V_{OUTM} \quad \text{(Eq. 12)}$$

$$V_{OUT,\ CM}=(V_{OUTP}+V_{OUTM})/2 \quad \text{(Eq. 13)}$$

$$H_{DIFF}(z)=V_{OUT,DIFF}/V_{IN,DIFF}=2*C_{IN}/((C_R+C_F)-(C_F-C_R)z^{-1}) \text{ for low frequency input } (V_{IN,\ 1}\sim V_{IN,\ 2}) \quad \text{(Eq. 14)}$$

$$H_{CM}(z)=V_{OUT,\ CM}/V_{IN,\ CM}=\text{Not Defined} \quad \text{(Eq.15)}$$

The common mode transfer function $H_{CM}(z)$ is not defined since there is no path for the input common mode to the output. Furthermore, there is no feedback of the output common mode voltage. In the presence of offsets, the output of the filter will saturate and the filter will no longer be able to perform its intended function.

FIG. **5** shows a circuit diagram of one embodiment of a switched capacitor filter with common mode feedback **500**. According to the teachings of the present invention, in one embodiment, it is determined that the common mode output signal can be fed back without using the operational amplifier. Thus the problem with the common-mode loop setting the overall bandwidth of the amplifier can be avoided. Since only a DC offset needs to be fed back, the switched capacitor circuit as shown in FIG. **5** can be used to nullify the charge introduced by the offset. In one embodiment, the switched capacitor circuit **500** includes a differential amplifier (also called main amplifier herein) **510** and an operational amplifier (also called error amplifier herein) **560**. The differential amplifier **510** includes input terminals **502**, **504**, and output terminal **532** which provides output voltage Voutm and output terminal **542** which provides output voltage Voutp, respectively. An input voltage Vinp **512** is coupled to a first terminal of a switch **513**. A second terminal of the switch **513** is coupled to a first terminal (also called first end) of a capacitor **514**. A second terminal (also called second end) of the capacitor **514** is coupled to a first terminal of a switch **516**. A second terminal of the switch **516** is coupled to the input terminal **502** of the differential amplifier **510**. As illustrated in FIG. **5**, another input voltage Vinm **522** is coupled to a first terminal of a switch **523**. A second terminal of the switch **523** is coupled to a first terminal (also called first end) of a capacitor **524**. A second terminal (also called second end) of the capacitor **524** is coupled to a first terminal of a switch **526**. A second terminal of the switch **526** is coupled to the input terminal **504** of the differential amplifier **510**. The second terminal of the capacitor **514** is also connected to a first terminal of a switch **515**. A second terminal of the switch **515** is connected to ground. Similarly, the second terminal of the capacitor **524** is also connected to a first terminal of a switch **525**. A second terminal of the switch **525** is connected to ground.

Referring again to FIG. **5**, the output terminal **532** is coupled to a first end of a capacitor **533** and to a first end of a switch **535**. A second end of the capacitor **533** is coupled to the input terminal **502**. A second end of the switch **535** is coupled to a first end of a capacitor **534**. The first end of capacitor **534** is also coupled to a first end of a switch **536**. A second end of the capacitor **534** is coupled to node **517** which is coupled to the second end of the capacitor **514** and the first end of the switch **516**. A second end of the switch **536** is coupled to the output terminal **542** of the differential amplifier **510**.

As shown in FIG. **5**, the output terminal **542** is coupled to a first end of a capacitor **543** and to a first end of a switch **545**. A second end of the capacitor **543** is coupled to the input terminal **504**. A second end of the switch **545** is coupled to a first end of a capacitor **544**. The first end of capacitor **544** is also coupled to a first end of a switch **546**. A second end of the capacitor **544** is coupled to node **527** which is coupled to the second end of the capacitor **524** and the first end of the switch **526**. A second end of the switch **546** is coupled to the output terminal **532** of the differential amplifier **510**.

In one embodiment, the output terminal **532** is coupled to a first terminal of a resistor **580**. A second terminal of the resistor **580** is coupled to node **570** which is connected to an input terminal of the error amplifier **560**. The output terminal **542** is coupled to a first terminal of a resistor **582**. A second terminal of the resistor **582** is also coupled to node **570**. The other input terminal of the error amplifier **560** is connected to a common mode reference voltage Vcm. The output terminal **586** of the error amplifier **560** is coupled to a first end of a switch **566**. The second end of the switch **566** is coupled to a first end of a capacitor **564** and a first end of a switch **562**. A second end of the switch **562** is coupled to the input terminal **502** of the differential amplifier **510**. A second end of the capacitor **564** is coupled to ground level. The output terminal **586** is also coupled to a first end of a switch **576**. A second end of the switch **576** is coupled to a first end of a capacitor **574** and a first end of a switch **572**. A second end of the switch **572** is coupled to the input terminal **504** of the differential amplifier **510**. A second end of the capacitor **574** is coupled to ground level.

In one embodiment, all the switches illustrated in FIG. **5** are turned on (closed) or turned off (open) in response to a corresponding control signal. For example, switches **513**, **523**, **515**, **525**, **535**, **545**, **562**, and **572** are operated in response to a control signal Θ_1. Switches **516**, **526**, **519**, **529**, **536**, **546**, **566**, and **576** are operated in response to a control signal Θ_2. In one embodiment, control signals Θ_1 and Θ_2 are non-overlapping clock signals. Switches illustrated in FIG. **5** can be any type of switching devices suitable for implementation in switched capacitor circuits in accordance with the teachings of the present invention. For example, various types of switching transistors can be used depending upon the particular applications or implementations of the present invention.

In one embodiment, when the control signal Θ_1 is at a first level (e.g., high logic level) and the control signal Θ_2 is at a second level (e.g., low logic level), switches **513**, **523**, **515**, **525**, **535**, **545**, **562**, and **572**, etc. are turned on or closed. When the control signal Θ_1 is at a second level (e.g., low logic level) and the control signal Θ_2 is at a first level (e.g., high logic level), switches **516**, **526**, **519**, **529**, **536**, **546**, **566**, and **576**, etc. are turned on or closed.

In operation, when switches **513** and **523** are turned on (closed), the Vinp input voltage is connected to the first terminal of the capacitor **514** and the Vinm input voltage is connected to the first terminal of the capacitor **524**, respectively. The second terminals of capacitors **514** and **524** are connected to ground. When switches **516**, **526**, **519**, and **529** are turned on (closed), the second terminal of capacitor **514** is connected to the input terminal **502** and the second terminal of capacitor **524** is connected to input terminal **504**, respectively. At this time, the Vinp input voltage is connected to the first terminal of capacitor **524** and the Vinm input voltage is connected to the first terminal of capacitor **514**, respectively. A common mode output voltage at node **570** is coupled to input terminal **552** of the error amplifier **560** while the input terminal **554** is coupled to a common mode reference voltage Vcm. The error amplifier **560**, in one embodiment, operates to compares the common mode output voltage at node **570** to the common mode reference voltage Vcm. When the switches **566** and **576** are closed, the output of the error amplifier **560** is coupled to capacitors **564** and **574**, respectively. When switches **562** and **572** are closed, capacitors **564** and **574** are coupled to the input terminals **502** and **504** of the main amplifier **510**, respectively. Accordingly, the switched capacitors **564** and **574** function to provide feedback of the common mode voltage for the differential amplifier **510**. Since the error amplifier **560** can have very low bandwidth and the feedback capacitors **564** and **574** can be made very small, the additional power consumption by the common mode feedback loop in the configuration illustrated in FIG. **5** will be negligible. In addition, the common mode sensing resistors **580** and **582** can be made very large to make their impact on the differential amplifier **510** negligible as well. The differential mode transfer function is unchanged. It can be appreciated and understood by one skilled in the art that, in accordance with the teachings of the present invention, the common mode output signal can be fed back without using the main amplifier **510**. Thus the problem with the common-mode loop setting the overall bandwidth of the main amplifier **510** can be avoided. Since only a DC offset needs to be fed back, the switched capacitor circuit as shown in FIG. **5** can be used to nullify the charge introduced by the offset.

FIG. **6** shows a circuit diagram of another embodiment of a switched capacitor filter with common mode feedback **600**, according to the teachings of the present invention. In one embodiment, the switched capacitor circuit **600** includes a differential amplifier (also called main amplifier herein) **610**, a pseudo-differential amplifier **660**, and an adder **690**. The differential amplifier **610** has two input terminals **602**, **604**, and output terminal **632** which provides output voltage Voutm and output terminal **642** which provides output voltage Voutp, respectively. In one embodiment, an input voltage Vinp **612** is coupled to a first terminal of a switch **613**. A second terminal of the switch **613** is coupled to a first terminal (also called first end) of a capacitor **614**. A second terminal (also called second end) of the capacitor **614** is coupled to a first terminal of a switch **616**. A second terminal of the switch **616** is coupled to the input terminal **602** of the differential amplifier **610**. In one embodiment, another input voltage Vinm **622** is coupled to a first terminal of a switch **623**. A second terminal of the switch **623** is coupled to a first terminal (also called first end) of a capacitor **624**. A second terminal (also called second end) of the capacitor **624** is coupled to a first terminal of a switch **626**. A second terminal of the switch **626** is coupled to the input terminal **604** of the differential amplifier **610**. The second terminal of the capacitor **614** is also connected to a first terminal of a switch **615**. A second terminal of the switch **615** is connected to ground. Similarly, the second terminal of the capacitor **624** is also connected to a first terminal of a switch **625**. A second terminal of the switch **625** is connected to ground.

In one embodiment, as shown in FIG. **6**, the output terminal **632** is coupled to a first end of a capacitor **633** and to a first end of a switch **635**. A second end of the capacitor **633** is coupled to the input terminal **602**. A second end of the switch **635** is coupled to a first end of a capacitor **634**. The first end of capacitor **634** is also coupled to a first end of a switch **636**. A second end of the capacitor **634** is coupled to node **617** which is coupled to the second end of the capacitor **614** and the first end of the switch **616**. A second end of the switch **636** is coupled to the output terminal **642** of the differential amplifier **610**.

In one embodiment, as shown in FIG. **6**, the output terminal **642** is coupled to a first end of a capacitor **643** and to a first end of a switch **645**. A second end of the capacitor **643** is coupled to the input terminal **604**. A second end of the switch **645** is coupled to a first end of a capacitor **644**. The first end of capacitor **644** is also coupled to a first end of a switch **646**. A second end of the capacitor **644** is coupled to node **627** which is coupled to the second end of the capacitor **624** and the first end of the switch **626**. A second end of the switch **646** is coupled to the output terminal **632** of the differential amplifier **610**.

Referring again to FIG. **6**, in one embodiment, the output terminals **632** and **642** are coupled to the negative inputs **652** and **655** of the pseudo-differential amplifier **660**, respectively. The positive inputs **653** and **654** of the pseudo-differential amplifier **660** are coupled to a common mode reference voltage Vcm. The output terminals **692** and **694** of the pseudo-differential amplifier **660** are coupled to the inputs of the adder **690**. In one embodiment, the output terminal **696** of the adder **690** is coupled to a first end of a switch **666**. The second end of the switch **666** is coupled to a first end of a capacitor **664** and a first end of a switch **662**. A second end of the switch **662** is coupled to the input terminal **602** of the differential amplifier **610**. A second end of the capacitor **664** is coupled to ground level. The output terminal **696** is also coupled to a first end of a switch **676**. A second end of the switch **676** is coupled to a first end of a capacitor **674** and a first end of a switch **672**. A second end of the switch **672** is coupled to the input terminal **604** of the differential amplifier **610**. A second end of the capacitor **674** is coupled to ground level.

In one embodiment, all the switches illustrated in FIG. **6** are turned on (closed) or turned off (open) in response to a corresponding control signal. For example, switches **613**, **623**, **615**, **625**, **635**, **645**, **662**, and **672** are operated in response to a control signal Θ_1. Switches **616**, **626**, **619**, **629**, **636**, **646**, **666**, and **676** are operated in response to a control signal Θ_2. In one embodiment, control signals Θ_1 and Θ_2 are non-overlapping clock signals. Switches illustrated in FIG. **6** can be any type of switching devices suitable for implementation in switched capacitor circuits in accordance with the teachings of the present invention. For example, various types of switching transistors can be used depending upon the particular applications or implementations of the present invention.

In one embodiment, when the control signal Θ_1 is at a first level (e.g., high logic level) and the control signal Θ_2 is at a second level (e.g., low logic level), switches **613**, **623**, **615**, **625**, **635**, **645**, **662**, and **672**, etc. are turned on or closed. When the control signal Θ_1 is at a second level (e.g., low logic level) and the control signal Θ_2 is at a first level (e.g., high logic level), switches **616**, **626**, **619**, **629**, **636**, **646**, **666**, and **676**, etc. are turned on or closed.

In operation, when switches **613** and **623** are turned on (closed), the Vinp input voltage is connected to the first terminal of the capacitor **614** and the Vinm input voltage is connected to the first terminal of the capacitor **624**, respectively. The second terminals of capacitors **614** and **624** are connected to ground. When switches **616**, **626**, **619**, and **629** are turned on (closed), the second terminal of capacitor **614** is connected to the input terminal **602** and the second terminal of capacitor **624** is connected to input terminal **604**, respectively. At this time, the Vinp input voltage is connected to the first terminal of capacitor **624** and the Vinm input voltage is connected to the first terminal of capacitor **614**, respectively. When the switches **666** and **676** are closed, the output of the adder **690** is coupled to capacitors **664** and **674**, respectively. When the switches **662** and **672** are closed, the capacitors **664** and **674** are coupled to the input terminals **602** and **604**, respectively. Accordingly, the switched capacitors **664** and **674** function to provide feedback of the common mode voltage for the differential amplifier **610**. Since the pseudo-differential amplifier **690** can have very low bandwidth and the feedback capacitors **664** and **674** can be made very small, the additional power consumption by the common mode feedback loop in the configuration illustrated in FIG. **6** will be negligible. The differential mode transfer function is unchanged.

It can be appreciated and understood by one skilled in the art that, in accordance with the teachings of the present invention, the common mode output signal can be fed back without using the main amplifier **610**. Thus the problem with the common-mode loop setting the overall bandwidth of the main amplifier **610** can be avoided. Since only a DC offset needs to be fed back, the switched capacitor circuit as shown in FIG. **6** can be used to nullify the charge introduced by the offset.

FIG. **7** shows a block diagram of a system **700** in which the teachings of the present invention are implemented. The system **700**, in one embodiment, may represent an example of a baseband channel circuitry employed in various wireless communication systems. As shown in FIG. **7**, the system **700** includes an anti-alias filter unit **710**, a switched capacitor filter unit **720**, and an analog-to-digital conversion (ADC) unit **730**. In one embodiment, the anti-alias filter unit **710** is a continuous time active-RC filter that is used to perform coarse filtering of the input signals (e.g., RF signals). In one embodiment, the switched capacitor filter unit **720** is configured as described above with respect to FIGS. **5** and **6**, in accordance with the teachings of the present invention. The switched capacitor filter unit **720** can be used to perform channel selection by filtering out nearby adjacent channels of the output signals from the anti-alias filter unit **710**. The ADC unit **730** is used to digitize the output analog signals generated by the switched capacitor filter unit **720** prior to the digital signal processing stage.

The invention has been described in conjunction with the preferred embodiment. It is evident that numerous alternatives, modifications, variations and uses will be apparent to those skilled in the art in light of the foregoing description.

What is claimed is:

1. An apparatus comprising:

a first amplifier including a first input, a second input, a first output, and a second output, the first input and the second input to receive a first input voltage and a second input voltage, respectively, and the first and second outputs to provide a first output voltage and a second output voltage, respectively;

a common mode feedback circuit having first and second switched capacitors coupled to provide a common mode feedback signal based on the first and second output voltages to the first and second inputs of first amplifier;

a third capacitor coupled between the first input and the first output of the first amplifier; and

a fourth capacitor coupled between the second input and the second output of the first amplifier.

2. The apparatus of claim **1** further including:

a fifth capacitor alternately coupled between the first input and the first output during a first period and coupled between the first input and the second output during a second period; and

a sixth capacitor alternately coupled between the second input and the second output during the first period and coupled between the second input and the first output during the second period.

3. An apparatus comprising:

a first amplifier including a first input, a second input, a first output, and a second output, the first input and the second input to receive a first input voltage and a second input voltage, respectively, and the first and second outputs to provide a first output voltage and a second output voltage, respectively; and

a common mode feedback circuit having first and second switched capacitors coupled to provide a common mode feedback signal based on the first and second output voltages to the first and second inputs of first amplifier,

wherein the first and second input voltages are alternately coupled to the first input and the second input of the first amplifier, based on first and second control signals.

4. The apparatus of claim **3** wherein the first and second control signals are non-overlapping clock signals.

5. An apparatus comprising:

a first amplifier including a first input, a second input, a first output, and a second output, the first input and the second input to receive a first input voltage and a second input voltage, respectively, and the first and second outputs to provide a first output voltage and a second output voltage, respectively; and

a common mode feedback circuit having first and second switched capacitors coupled to provide a common mode feedback signal based on the first and second output voltages to the first and second inputs of first amplifier, wherein the common mode feedback circuit includes a component to generate the common mode feedback signal based on the first output voltage, the second output voltage, and a common mode reference voltage,

wherein the common mode feedback signal is coupled to the first capacitor via a first switch and coupled to the second capacitor via a second switch, and

wherein the first capacitor is coupled to the first input via a third switch and the second capacitor is coupled to the second input via a fourth switch.

6. The apparatus of claim **5** wherein the third and fourth switches are turned on and off based upon a value of a first control signal.

7. The apparatus of claim **6** wherein the first and second switches are turned on and off based upon a value of a second control signal.

8. An apparatus comprising:

a first amplifier including a first input, a second input, a first output, and a second output, the first input and the second input to receive a first input voltage and a second input voltage, respectively, and the first and second outputs to provide a first output voltage and a second output voltage, respectively; and

a common mode feedback circuit having first and second switched capacitors coupled to provide a common mode feedback signal based on the first and second output voltages to the first and second inputs of first amplifier,

wherein the common mode feedback circuit includes a component to generate the common mode feedback signal based on the first output voltage, the second output voltage, and a common mode reference voltage, and

wherein the component includes a second amplifier having a first input and a second input, the first input being coupled to a common mode output voltage corresponding to a low-pass filtered version of the difference between the first output voltage and the second output voltage, the second input being coupled to the common mode reference voltage.

9. The apparatus of claim **8** wherein the second amplifier is an error amplifier.

10. An apparatus comprising:

a first amplifier including a first input, a second input, a first output, and a second output, the first input and the second input to receive a first input voltage and a second input voltage, respectively, and the first and second outputs to provide a first output voltage and a second output voltage, respectively; and

a common mode feedback circuit having first and second switched capacitors coupled to provide a common mode feedback signal based on the first and second output voltages to the first and second inputs of first amplifier,

wherein the common mode feedback circuit includes a component to generate the common mode feedback signal based on the first output voltage, the second output voltage, and a common mode reference voltage, and

wherein the component includes:

a pseudo-differential amplifier having first and second negative inputs, first and second positive inputs, the first and second negative inputs being coupled to the first and second output voltages, respectively, the first and second positive inputs being coupled to a common reference voltage; and

an adder coupled to generate the common mode feedback signal based on first and second outputs of the pseudo-differential amplifier.

11. A method comprising:

receiving a first output voltage and a second output voltage from a first amplifier;

coupling the first and second output voltages to a first negative input and second negative input of a pseudo-differential amplifier, respectively;

generating a common mode feedback signal based on a first output and a second output of the pseudo-differential amplifier; and

providing the common mode feedback signal to a first input and a second input of the first amplifier via a first switched capacitor and a second switched capacitor, respectively.

12. A method comprising:

receiving a first output voltage and a second output voltage from a first amplifier;

generating a common mode feedback signal based on the first and second output voltages; and

providing the common mode feedback signal to first and second inputs of the first amplifier via first and second switched capacitors, respectively,

wherein generating the common mode feedback signal including:

coupling a third voltage corresponding to a low-pass filtered version of the difference between the first and second output voltages to a first input of a second amplifier;

coupling a second input of the second amplifier to a common mode reference voltage; and

generating the common mode feedback signal based on the difference between the third voltage and the common mode reference voltage.

13. A method comprising:

receiving a first output voltage and a second output voltage from a first amplifier;

generating a common mode feedback signal based on the first and second output voltages; and

providing the common mode feedback signal to first and second inputs of the first amplifier via first and second switched capacitors, respectively, including:

coupling the common mode feedback signal to first and second capacitors via first and second switches, respectively; and

coupling first and second capacitors to first and second capacitors to first and second inputs of the first amplifier via third and fourth switches.

14. The method of claim **13** wherein the third and fourth switches are turned on to couple the first and second capacitors to first and second inputs of the first amplifier, respectively, and turned off to decouple the first and second capacitors from first and second inputs of the first amplifier, respectively.

15. The method of claim **14** wherein the third and fourth switches are turned on and off based on the value of a first control signal.

16. The method of claim **15** wherein the first and second switches are turned on to couple the common mode feedback signal to the first and second capacitors, respectively, and turned off to decouple the common mode feedback signal from the first and second capacitors, respectively.

17. The method of claim **16** wherein the first and second switches are turned on and off based on the value of a second control signal.

18. The method of claim 17 wherein first and second control signals are non-overlapping clock signals.

19. A system comprising:

a first filter to perform coarse filtering of input signals;

a second filter coupled to the first filter to perform channel selection of the output signals generated by the first filter, the second filter including:

a first amplifier including a first input, a second input, a first output, and a second output, the first input and the second input to receive a first input voltage and a second input voltage, respectively, and the first and second outputs to provide a first output voltage and a second output voltage, respectively; and

a common mode feedback circuit having first and second switched capacitors coupled to provide a common made feedback signal based on the first and second output voltages to the first and second inputs of first amplifier; and

an analog-to-digital converter (ADC) coupled to the second filter to digitize the output of the second filter.

20. The system of claim 19 further including:

a third capacitor coupled between the first input and the first output of the first amplifier; and

a fourth capacitor coupled between the second input and the second output of the first amplifier.

21. The system of claim 20 further including:

a fifth capacitor alternately coupled between the first input and the first output during a first period and coupled between the first input and the second output during a second period; and

a sixth capacitor alternately coupled between the second input and the second output during the first period and coupled between the second input and the first output during the second period.

22. The system of claim 19 wherein the first and second input voltages are alternately coupled to the first input and the second input of the first amplifier, based on first and second control signals.

23. The system of claim 19 wherein the common mode feedback circuit includes a component to generate the common mode feedback signal based on the first output voltage, the second output voltage, and a common mode reference voltage.

24. The system of claim 23 wherein the component includes:

a second amplifier having a first input and a second input, the first input being coupled to a common mode output voltage corresponding to a low-pass filtered version of the difference between the first output voltage and the second output voltage, the second input being coupled to the common mode reference voltage.

25. The system of claim 24 wherein the second amplifier is an error amplifier.

26. The system of claim 23 wherein the component includes:

a pseudo-differential amplifier having first and second negative inputs, first and second positive inputs, the first and second negative inputs being coupled to the first and second output voltages, respectively, the first and second positive inputs being coupled to a common reference voltage; and

an adder coupled to generate the common mode feedback signal based on first and second outputs of the pseudo-differential amplifier.

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